

Tugas Lab. Algo Pertemuan 12

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Kelas : TI-25-PA2

Github: <https://github.com/Fathur123-TI/LabAlgoritmaSemester2.git>

```
1  #include <iostream>
2  #include <limits>
3  using namespace std;
4
5  const int V = 6;
6
7  int minDistance(int dist[], bool visited[]) {
8      int min = INT_MAX;
9      int minIndex = -1;
10
11     for (int i = 0; i < V; i++) {
12         if (!visited[i] && dist[i] < min) {
13             min = dist[i];
14             minIndex = i;
15         }
16     }
17     return minIndex;
18 }
19
20 void printPath(int parent[], int j) {
21     if (parent[j] == -1) {
22         cout << j + 1;
23         return;
24     }
25
26     printPath(parent, parent[j]);
27     cout << " -> " << j + 1;
28 }
29
30 void dijkstra(int graph[V][V], int src, int dest) {
31     int dist[V];
32     bool visited[V];
33     int parent[V];
34
35     for (int i = 0; i < V; i++) {
36         dist[i] = INT_MAX;
37         visited[i] = false;
38         parent[i] = -1;
39     }
40
41     dist[src] = 0;
42
43     for (int i = 0; i < V - 1; i++) {
44         int u = minDistance(dist, visited);
45
46         if (u == -1)
47             break;
48
49         visited[u] = true;
50
51         for (int v = 0; v < V; v++) {
52             if (!visited[v] &&
53                 graph[u][v] != 0 &&
54                 dist[u] != INT_MAX &&
55                 dist[u] + graph[u][v] < dist[v]) {
56                 dist[v] = dist[u] + graph[u][v];
57                 parent[v] = u;
58             }
59         }
60     }
61
62     cout << "Shortest distance from node "
63          << src + 1 << " to node "
64          << dest + 1 << " = "
```

```

64
65     cout << "Shortest distance from node "
66         << src + 1 << " to node "
67         << dest + 1 << " = "
68         << dist[dest] << endl;
69
70     cout << "Path : ";
71     printPath(parent, dest);
72     cout << endl;
73 }
74
75 int main() {
76
77     int graph[V][V] = {
78         {0,4,2,0,0,0},
79         {4,0,1,5,0,0},
80         {2,1,0,8,10,0},
81         {0,5,8,0,2,6},
82         {0,0,10,2,0,3},
83         {0,0,0,6,3,0}
84     };
85
86     dijkstra(graph, 0, 4);
87
88     return 0;
89 }

```

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```

Shortest distance from node 1 to node 5 = 10
Path : 1 -> 3 -> 2 -> 4 -> 5

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Process exited after 0.05516 seconds with return value 0
Press any key to continue . . . |

```